EFU-OXYGEN STANDARD v1.0

Flux-Based Human-Equivalent Oxygen Accounting Framework

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ISO compatibility statement

The EFU-Oxygen Standard v1.0 is **not** an ISO standard, not a certification scheme, and not a conformity assessment tool. It is **conceptually alignable** with ISO 14001 environmental management systems (particularly where dissolved oxygen, BOD/COD and combustion processes are relevant environmental aspects), ISO 14046 (water footprint), and ISO 14064 (greenhouse gas quantification), but does not imply any formal endorsement, co-development, or approval by ISO or its technical committees.

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1. Executive Summary

The **EFU-Oxygen Standard v1.0** defines a flux-based, human-equivalent metric for measuring molecular oxygen (O_2) flows in anthropogenic and ecological systems, using average adult respiratory oxygen consumption as a reference baseline.

Normative reference value (modelling convention):

- **1 EFU-O_2 = 230 kg O_2/year/person**
  - A representative mid-range value within the ~200–260 kg O_2/year interval observed for adult oxygen consumption, used as a comparative modelling baseline.

Approximate equivalents:

- ≈ 300–350 mL O_2/min (including typical daily activity)
- ≈ 0.6–0.7 kg O_2/day
- ≈ 4.4–4.6 × 10^2 L O_2/day at STP (0 °C, 1 atm)
- ≈ 6.0–6.5 × 10^2 L O_2/day at body temperature and humidity (BTPS)

These are **modelling conventions**, not new biological constants; they provide a human-scale reference against which combustion systems, industrial processes, aquatic dissolved oxygen dynamics, and ecosystem services can be expressed in **EFU-O_2 units** (human-equivalent oxygen flux).

As a **supplementary environmental performance indicator**, EFU-O_2 can support:

- Identification of environmental aspects under ISO 14001 (e.g. river hypoxia, combustion-related O_2 demand).archive.
- Enhanced interpretation of ISO 14046 water-footprint assessments where dissolved oxygen and eutrophication are critical.
- Parallel oxygen-footprint perspectives alongside ISO 14064 greenhouse-gas inventories.

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2. Scientific Foundations

2.1 Human Oxygen Metabolism

Human tissues require continuous oxygen supply for aerobic metabolism and ATP production:

$$\boxed{\mathbf{C6H12O6 + 6O2 \rightarrow 6CO2 + 6H2O + energy\ (ATP)}}$$

Basal metabolic rate (BMR) represents resting energy expenditure and is directly linked to oxygen uptake. Adult BMR values commonly fall in the ~1 300–1 800 kcal/day range, corresponding to resting oxygen uptakes of roughly 200–250 mL O_2/min.

Including everyday activity (Physical Activity Level ≈ 1.4–1.7), average daily oxygen consumption shifts into the ~300–350 mL O_2/min range, which equates to ≈0.6–0.7 kg O_2/day or approximately 200–260 kg O_2/year for an average adult.

EFU-O_2 reference definition (conventional baseline):

- **1 EFU-O_2 = 230 kg O_2/year/person**
  - A representative mid-point within the 200–260 kg/year adult interval.
  - Associated approximate equivalents:
    - ≈ 300–350 mL O_2/min (activity-adjusted),
    - ≈ 0.6–0.7 kg O_2/day,
    - ≈ 4.4–4.6 × 10^2 L O_2/day at STP,
    - ≈ 6.0–6.5 × 10^2 L O_2/day at BTPS.

Volumetric conversion at STP (0 °C, 1 atm)

- Molecular weight O_2: 32 g/mol
- Molar volume at STP: 22.4 L/mol
- Density at STP: ρ ≈ 1.429 kg/m³

$$\boxed{\mathbf{230kg/e'v = 230000g \approx 1,429230000 \approx 1,61 \times 105L/e'v}}$$

$$\boxed{\mathbf{\Rightarrow \approx 440-460L/nap\ STPn \approx 600-650L/nap\ BTPSen}}$$

The EFU-O_2 reference is therefore a **human-equivalent oxygen flux**, anchored in plausible adult physiology yet explicitly defined as a **modelling convention** rather than a new SI base unit.

2.2 Atmospheric Oxygen Reservoir and Global Fluxes

- Total atmospheric O_2 mass: ~1.2×10^{18} kg.
- Volume fraction of O_2: ~20.95%.
- Characteristic residence time: several thousand years.

Annual fluxes (order of magnitude):

- Gross photosynthetic O_2 production: ~3×10^{14} kg O_2/year.
- Global respiration and oxidative losses: ~3×10^{14} kg O_2/year.
- Observed atmospheric O_2 decline: ~0.001–0.002% per year, cumulating to a few tenths of a percent decrease since the late 20th century, consistent with fossil-fuel driven CO_2 increases.

Anthropogenic O_2 consumption

- Global population: ~8×10^9.
- Human respiration (at EFU-O_2 baseline):

$$\boxed{8 \times 109 \times 230\text{kg/e´v} = 1,84 \times 1012\text{kg O2/e´v}}$$

- Fossil fuel combustion: ~35 Gt CO_2/year → ≈35 Gt O_2/year consumed (near-unity mass ratio for global fuel mix).nrdc+1
- Biomass burning: order of 10^{13} kg O_2/year.

Expressed in EFU-O_2:

- Human respiration: ~8.0×10^9 EFU-O_2.
- Fossil fuels: ~1.5×10^{11} EFU-O_2.
- Biomass burning: ~4.3×10^{10} EFU-O_2.

Total anthropogenic O_2 consumption (metabolic + combustion) is ≈2.0×10^{11} EFU-O_2, i.e. roughly 25 EFU-O_2/person/year globally.

Interpretation

- Absolute atmospheric O_2 depletion remains small relative to the huge reservoir and does **not** currently constitute an acute oxygen crisis.
- The decline pattern is, however, a robust **physical fingerprint** of fossil-fuel-driven oxidation processes and can be quantitatively described using EFU-O_2 alongside CO_2 metrics.

2.3 Combustion Stoichiometry and O_2:CO_2 Ratios

Generic hydrocarbon combustion:

$$\boxed{\mathbf{C}n\mathbf{H}m + (n + 4m)\mathbf{O2} \rightarrow n\mathbf{CO2} + 2m\mathbf{H2O}}$$

Examples:

- Coal (≈C): $C + O_2 \rightarrow CO_2$ (molar ratio 1:1; mass ratio 32:44).
- Methane (CH_4): $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ (molar ratio 2:1; mass ratio 64:44).

For realistic fossil fuel mixes, a typical **average** is close to ~1.0 kg O_2 consumed per 1.0 kg CO_2 emitted, recognising fuel-specific deviations.

This underpins the mapping between EFU-C and EFU-O_2 (Section 5.1).

2.4 Dissolved Oxygen in Aquatic Systems

- Saturation DO in freshwater at 20 °C, 1 atm: ≈8–9 mg O_2/L.
- Hypoxic threshold: <2 mg O_2/L.
- Anoxic: <0.5 mg O_2/L.

Biodegradation of organic matter produces biochemical oxygen demand (BOD):

- Complete degradation of 1 kg organic matter typically requires ~1–2 kg O_2 (order of magnitude).

EFU-O_2 allows O_2 demand and deficits in rivers, lakes, and coastal zones to be expressed as **human-equivalent oxygen consumption**, making hypoxic and anoxic risks more communicable.

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3. Conversion Formulas

3.1 Mass to EFU-O_2

$$\boxed{\mathbf{EFU\text{-}O2 = 230kg\ O2/e´v}}$$

3.2 Volume to EFU-O_2 (Gas, STP)

With ρ ≈ 1.429 kg/m³ at STP:

$$\boxed{\mathbf{EFU\text{-}O2 = 1{,}61 \times 105L\ O2/e´v\ (STP)}}$$
$$\boxed{\mathbf{EFU\text{-}O2 = 161m3\ O2/e´v\ (STP)}}$$

(Example unchanged from previous draft.)

3.3 Fuel-Specific O_2 Demand (Approximate)

Same table and explanations as in the Hungarian version: approximate engineering factors assuming complete combustion and typical compositions.

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4. Implementation Guidelines

Same structure as previously drafted:

- 4.1 Minimum data requirements (atmospheric and aquatic).
- 4.2 Oxygen Impact Statement template.

No changes in content needed, only the ISO-compatibility context now frames their use.

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5. Integration with Other EFU Standards

5.1 EFU-Carbon (EFU-C)

Include the relation:

1 EFU-C≈1.14 EFU-O21\,\text{EFU-C} \approx 1.14\,\text{EFU-O}_21EFU-C≈1.14EFU-O2

as already derived, with the note that this assumes fossil-fuel combustion and the EFU-C baseline of 263 kg CO_2/year/person.

5.2 EFU-Energy (EFU-H)

Retain the EFU-H linkage and the per-TJ O_2 intensity table, with the clarifying sentence that hydrogen can show lower O_2 demand per useful unit of energy than coal or oil systems, while non-combustion energy sources have effectively zero atmospheric O_2 demand.

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6. Relationship with ISO Standards (Informative)

The EFU-Oxygen Standard v1.0 is a stand-alone, research-focused framework that can be **conceptually aligned** with several existing ISO standards without becoming a formal part of them:

- **ISO 14001 – Environmental management systems**

EFU-O_2 can support the identification and monitoring of environmental aspects such as aquatic oxygen depletion (BOD/COD-driven DO decline) and combustion-related oxygen use in energy systems and industrial processes, in line with ISO 14001's environmental aspects/impacts logic.ocw.

- **ISO 14046 – Water footprint**

  EFU-O_2 can complement water-footprint indicators by expressing dissolved oxygen deficits and eutrophication-related impacts in human-equivalent terms, particularly where aquatic ecosystem capacity and hypoxia risk are critical.

- **ISO 14064 – Greenhouse gases**

  EFU-O_2 provides an **oxygen-footprint** perspective parallel to CO_2 inventories, using the approximate 1:1 O_2:CO_2 mass relationship, and can be mapped directly to EFU-C-based GHG accounting.

These linkages indicate **conceptual compatibility**, not formal collaboration or endorsement. EFU-O_2 usage never replaces the need to comply with the specific requirements of these ISO standards, and does not constitute ISO-conform certification.

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7. Status, Scope, and Limitations

The EFU-Oxygen Standard v1.0 is a:

- **Pre-standard technical architecture** within the EFU biophysical accounting framework,
- **Research-grade experimental metric**,
- **Not** an ISO standard, **not** a regulatory scheme, and **not** a certified audit protocol.

Intended use:

- Scientific research.
- Methodological development.
- Retrospective analysis of completed projects.
- Controlled pilot implementations (e.g. as a supplementary indicator within ISO 14001/14046 environmental management or reporting systems).

Any use of EFU-O_2 outputs for formal policy, regulatory or certification purposes requires **independent validation, peer review, and dedicated adoption processes** (e.g. via national standards bodies, ISO/TC or ISO/IEC technical committees), which lie beyond the scope of this document.

All numerical values are presented as **order-of-magnitude or representative baselines**, explicitly open to refinement as physiological, atmospheric, and ecological datasets improve.

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8. Acknowledgements and AI Disclosure

- Primary conceptual development, system design, and scientific validation: **István Simor**.

AI tools acted as **cognitive and analytical supports**, not as autonomous authors or validators. Full responsibility for the conceptual architecture, system logic, and scientific content rests with the human author.

ISO disclaimer

The term “ISO” in this document is used solely to reference relevant international standards as contextual anchors. The EFU-Oxygen Standard has **not** been reviewed, approved, or adopted by ISO or any ISO/IEC technical committee; ISO-compatibility is strictly conceptual and methodological at this stage.